Yang–Mills Grassroots Formalization Notes

Mettapedia autoformalization notes

May 15, 2026

1 Scope

This note records first-party Lean work toward a Yang–Mills existence and mass gap formalization. The current target is deliberately modest: establish a small spectral mass-gap audit layer that future classical-gauge and QFT constructions must satisfy.

The Clay formulation says, in spectral terms, that the Hamiltonian has no spectrum in an interval $(0, \Delta)$ above the vacuum. The present Lean work does not construct a Yang–Mills theory, a Hilbert space, a Hamiltonian, or the Wightman/Osterwalder–Schrader infrastructure. It only fixes the bounded Hamiltonian gap predicates and proves basic consequences that should remain valid under later strengthening.

2 Lean Layer

The first module is

`Mettapedia/QuantumTheory/YangMills/MassGap.lean`.

It introduces:

- `Spacetime` as $\mathbb{R}^4$ and `Space` as $\mathbb{R}^3$, matching the coordinate types used by the existing reference statement;
- `LinearOperator`, the bounded real-linear operator surface used by Mathlib's current spectrum API;
- `SpectrumAvoidsOpenInterval`, a direct predicate for absence of spectrum in an interval;
- `HasSpectralMassGap`, the positive spectral window $0 < \Delta$ plus absence of spectrum in $(0, \Delta)$;
- `HasPositiveEnergySpectrum` and `HasVacuumEnergyZero` as separate spectral audit assumptions;
- `HasFirstPositiveSpectralValue`, the bounded-operator audit predicate for a first positive spectral excitation m;
- `PositiveSpectrumSet` and `PositiveSpectralInf` as positive-spectrum order audit surfaces;
- `SpectralGapSet`, `SpectralGapSup`, and `HasFiniteSpectralMass` as finite-gap audit surfaces;
- `HasQuadraticMassGap`, the Hilbert-space quadratic-form version often used as a proof obligation before invoking spectral machinery.

3 Proved Consequences

The module proves that spectral gaps are downward closed:

`HasSpectralMassGap.mono`.

Thus any proposed proof that produces a gap Δ also produces every smaller positive gap. This is a useful sanity check for future gap claims and for any definition of the mass as a supremum of admissible gaps.

It also proves a pointwise obstruction:

`not_hasSpectralMassGap_of_spectrum_in_gap`.

A single spectral value E with $0 < E < \Delta$ rules out Δ as a valid spectral mass gap. This is the Yang–Mills analogue of a concrete audit target: any future construction must explain why the Hamiltonian has no such low positive spectral values.

The same exclusion is now available in threshold and set-subset forms:

```
HasSpectralMassGap.gap_le_of_pos_spectrum
HasSpectralMassGap.spectrum_subset_nonpos_or_gap
```

Thus a positive spectral value of a Hamiltonian with gap Δ must lie at least at Δ, and the whole spectrum is confined to $(-\infty, 0] \cup [\Delta, \infty)$. The companion lemma

`HasSpectralMassGap.not_spectrum_of_pos_lt`

gives the direct operational exclusion of any $E \in (0, \Delta)$. Lean now also proves the exact converse characterization

`hasSpectralMassGap_iff_positive_and_spectrum_subset_nonpos_or_gap`.

So, on this bounded-operator audit surface, a positive Δ is a spectral mass gap exactly when the spectrum lies in $(-\infty, 0] \cup [\Delta, \infty)$.

When combined with positive energy, the gap splits the spectrum:

`HasSpectralMassGap.eq_zero_or_gap_le_of_positive_energy`.

Every spectral value is either 0 or at least Δ. This cleanly separates the vacuum-energy condition from the low-excitation exclusion. There are also theorem-level forms for non-vacuum spectral values and for the whole positive-energy spectrum:

```
HasSpectralMassGap.gap_le_of_positive_energy_of_ne_zero
HasSpectralMassGap.spectrum_subset_vacuum_or_gap_of_positive_energy
```

The same positive-energy split is now available as an exact iff:

`hasSpectralMassGap_iff_positive_and_spectrum_subset_vacuum_or_gap_of_positive_energy`.

Thus, once positive energy is fixed, the mass-gap predicate is interchangeable with the statement that every spectral value is either the vacuum value 0 or lies at least at Δ.

The first-positive-spectral-value interface packages the expected "first excitation" target without building the underlying Yang–Mills Hamiltonian. If m is the first positive spectral value, then lower positive spectral values are excluded and the whole spectrum lies in $(-\infty, 0] \cup [m, \infty)$:

```
HasFirstPositiveSpectralValue.unique
HasFirstPositiveSpectralValue.not_spectrum_of_pos_lt
HasFirstPositiveSpectralValue.spectrum_subset_nonpos_or_first
HasFirstPositiveSpectralValue.spectrum_subset_vacuum_or_first_of_positiveEnergy
```

The uniqueness lemma records that two supplied first positive spectral values for the same Hamiltonian must be equal. With positive energy, the nonpositive side of the split tightens to the vacuum alternative $E = 0$. The same predicate is also equivalent to the order-theoretic statement that m is the least element of the positive spectrum:

```
HasFirstPositiveSpectralValue.isLeast_positiveSpectrumSet
hasFirstPositiveSpectralValue_iff_isLeast_positiveSpectrumSet
HasFirstPositiveSpectralValue.positiveSpectralInf_eq
```

This gives the parallel audit value `PositiveSpectralInf` $A = m$ whenever the first positive spectral value exists.

The bridge back to the gap predicate is exact:

`hasFirstPositiveSpectralValue_iff_spectrum_mem_and_hasSpectralMassGap`.

Equivalently, a candidate m is the first positive spectral value precisely when m is actually in the spectrum and m itself is an admitted spectral mass gap. The forward direction is also available as

`HasSpectralMassGap.hasFirstPositiveSpectralValue_of_spectrum_mem`.

The same assumption exactly identifies the admitted spectral gaps:

```
HasFirstPositiveSpectralValue.hasSpectralMassGap_of_le
HasFirstPositiveSpectralValue.hasSpectralMassGap_iff
HasFirstPositiveSpectralValue.spectralGapSet_eq_Ioc
```

Thus the bounded audit layer proves `SpectralGapSet` $A = (0, m]$. It also supplies finite spectral mass and collapses the supremum audit to the first excitation:

```
HasFirstPositiveSpectralValue.hasFiniteSpectralMass
HasFirstPositiveSpectralValue.spectralGapSup_eq
HasFirstPositiveSpectralValue.spectralGapSup_eq_positiveSpectralInf
```

These are not Yang–Mills construction theorems; they are downstream checks that any proposed construction of a Hamiltonian with a first positive spectral value must satisfy.

The finite-mass audit predicate is linked to actual positive spectrum by

`hasFiniteSpectralMass_of_pos_spectrum`.

A single positive spectral value gives an upper bound for every admissible gap. The boundedness surface is now exposed directly as

`hasFiniteSpectralMass_iff_bddAbove_spectralGapSet`.

Thus the finite-mass predicate is exactly the assertion that the admitted gap set is bounded above, with the definition's positive bound recovered by taking the maximum of any upper bound and 1. The new supremum audit value

`SpectralGapSup`

is deliberately paired with the hypotheses that make it meaningful. If the gap set is bounded above and nonempty, then

`spectralGapSup_isLUB_of_hasFiniteSpectralMass_of_gap`

proves that `SpectralGapSup` is the least upper bound of all admitted gaps. The pointwise lower and positivity forms are

```
HasSpectralMassGap.le_spectralGapSup_of_hasFiniteSpectralMass
spectralGapSup_pos_of_hasFiniteSpectralMass_of_gap
```

and a positive spectral value bounds the supremum by

`spectralGapSup_le_of_pos_spectrum_of_gap`.

This is the supremum-level audit version of the elementary fact that every admitted gap must lie below any positive spectral value. Conversely,

`not_hasFiniteSpectralMass_of_arbitrarily_large_gaps`

shows that admitted gaps above every proposed bound rule out finite spectral mass. The earlier special case

`not_hasFiniteSpectralMass_of_all_positive_gaps`

records that if every positive window is accepted as a gap, then no finite upper bound on gaps can exist. This distinguishes a genuine finite first excitation from the degenerate situation where the bounded audit surface sees no positive spectrum at all.

Finally, the quadratic-form gap has the same downward closure:

`HasQuadraticMassGap.mono`.

This proof uses only nonnegativity of $\langle \psi, \psi \rangle$, so it is a stable preliminary lemma before any later unbounded-operator or spectral theorem bridge is added.

4 Exponential Clustering Audit

The next module is

`Mettapedia/QuantumTheory/YangMills/Clustering.lean`.

It isolates the Clay clustering consequence without pretending to prove the analytic mass-gap-to-clustering theorem. A two-point spatial correlation is represented abstractly as

$$\texttt{SpatialCorrelation} = \texttt{Space} \to \texttt{Space} \to \mathbb{R}.$$

For such a correlation kernel, the module defines:

- `HasExponentialClusteringAtRate`, the fixed-rate estimate $|C(x, y)| \leq \exp(-c\, d(x, y))$ beyond some radius;
- `HasExponentialClusteringUpTo`, the Clay-style all-rates-below-gap statement, requiring the fixed-rate estimate for every $0 < c < \Delta$;

- `HasArbitrarilyLargeClusteringViolation`, a concrete obstruction saying that every proposed cutoff radius has a farther counterexample to the fixed-rate exponential bound;
- `SpectralGapClusteringBridge`, an abstract bridge asserting that every spectral gap for a Hamiltonian implies clustering of the supplied correlation kernel up to that gap.

The basic monotonicity and projection lemmas are now formalized:

```
HasExponentialClusteringUpTo.at_rate
HasExponentialClusteringUpTo.mono
SpectralGapClusteringBridge.clusteringUpTo_of_gap
SpectralGapClusteringBridge.clusteringAtRate_of_gap
```

Thus, once a proof supplies a spectral-gap-to-clustering bridge, every proposed gap Δ produces each concrete clustering estimate with $0 < c < \Delta$. This deliberately separates the analytic theorem from the bookkeeping that future route audits will need.

The first crux-style consequence is also formalized:

```
not_hasExponentialClusteringAtRate_of_arbitrarilyLarge_violation
not_hasExponentialClusteringUpTo_of_rate_failure
not_hasSpectralMassGap_of_clustering_rate_failure
not_hasSpectralMassGap_of_arbitrarilyLarge_clustering_violation
spectralGap_le_of_clustering_rate_failure
spectralGap_le_of_arbitrarilyLarge_clustering_violation
spectralGapSup_le_of_clustering_rate_failure_of_gap
not_exists_hasSpectralMassGap_of_all_positive_clustering_rate_failure
not_exists_hasSpectralMassGap_of_all_positive_arbitrarilyLarge_clustering_violation
HasPositiveLongRangeLowerBound.arbitrarilyLargeClusteringViolation
not_exists_hasSpectralMassGap_of_positiveLongRangeLowerBound
```

Under a supplied `SpectralGapClusteringBridge`, failure of the exponential clustering estimate at any rate $0 < c < \Delta$ rules out Δ as a spectral mass gap for that Hamiltonian. This is not a global Yang–Mills blocker; it is a reusable audit surface for the manuscript's route from polymer/tree decay to exponential clustering and then to the OS Hamiltonian gap. Any later proof path that claims such a bridge must now provide the fixed-rate clustering estimates explicitly, or it exposes a concrete rate-level counterexample target.

The same obstruction is now available in quantitative threshold form. A single failed positive clustering rate c upper-bounds every bridged spectral gap:

$$\texttt{HasSpectralMassGap}\ A\ \Delta \quad \Longrightarrow \quad \Delta \leq c.$$

If at least one gap is admitted, the audited supremum `SpectralGapSup` is also bounded by c. Finally, if the supplied correlation fails the fixed-rate exponential estimate at every positive rate, Lean proves that there is no positive spectral gap at all for that bridged route. This converts a family of long-range correlation counterexamples into a direct no-gap criterion on the abstract Hamiltonian surface.

The newest concrete route-facing form is

$$\texttt{HasPositiveLongRangeLowerBound}\ \mathrm{corr}\ \varepsilon.$$

It says that for some $\varepsilon > 0$, beyond every radius there are two points whose absolute correlation is still at least ε. Lean proves that this lower bound beats every positive exponential decay rate:

`HasPositiveLongRangeLowerBound.arbitrarilyLargeClusteringViolation`.

Combining this with the spectral-gap-to-clustering bridge gives

`not_exists_hasSpectralMassGap_of_positiveLongRangeLowerBound`.

Thus a persistent nonzero long-range tail in the proposed two-point function is not merely a missing estimate; on this bridge surface it directly rules out all positive spectral gaps. This provides a concrete target for Yang–Mills route audits: either prove genuine exponential decay of the centered correlation functions, or avoid any lower-bound tail that persists uniformly at arbitrarily large separations.

5 RG Bootstrap Audit

The next route-facing module is

`Mettapedia/QuantumTheory/YangMills/RGBootstrap.lean`.

This file targets the manuscript's multiscale RG contraction claim in a narrow, checkable way. It does not prove the polymer RG step. Instead, it isolates the finite arithmetic and recurrence facts that any later formalized RG proof must instantiate.

The module defines

$$\texttt{irrelevantScale}\ b\ d_{\max} = b^{3-d_{\max}}$$

using an integer exponent, together with the predicates `HasRGContraction` and `HasExtendedExtractionContracti`
It also defines `AffineRGStepBound`, the recurrence surface

$$a_{n+1} \le \kappa a_n + \varepsilon.$$

The basic finite bootstrap lemma is

`AffineRGStepBound.bound_by_invariant_interval`.

If $a_0 \le B$, $\kappa \ge 0$, and $\varepsilon \le (1-\kappa)B$, then every finite iterate satisfies $a_n \le B$. This is the algebraic core of a smallness bootstrap once the analytic RG estimates have supplied a one-step recurrence.

The same module now also exposes the exact finite-iterate envelope

$$\texttt{affineRGEnvelope}\ \kappa\ \varepsilon\ a_0,$$

with theorem

`AffineRGStepBound.le_affineRGEnvelope`.

Thus any sequence satisfying the one-step affine bound is dominated at every finite scale by the recurrence $x_{n+1} = \kappa x_n + \varepsilon$ started at a_0. In the homogeneous case, Lean proves

`affineRGEnvelope_zero_error`

and the resulting wrapper

`AffineRGStepBound.bound_by_geometric_of_zero_error`,

so the envelope reduces exactly to $x_n = \kappa^n a_0$. This keeps the finite-scale bookkeeping proved independently of the still-open analytic RG step estimate.

The numerical extended-extraction audit is exact:

```
irrelevantScale_two_sixteen
irrelevantScale_two_four
irrelevantScale_two_fourteen
irrelevantScale_two_fifteen
irrelevantScale_eq_inv_pow_of_three_le
rgContraction_iff_constant_lt_pow_of_pos_of_three_le
rgContraction_forces_constant_lt_pow_of_pos_of_three_le
not_rgContraction_2224_of_pow_le_of_pos_of_three_le
irrelevantScale_two_eq_inv_pow_of_three_le
rgContraction_two_iff_constant_lt_pow_of_three_le
rgContraction_two_forces_constant_lt_pow_of_three_le
rgContraction_2224_two_sixteen
rgContraction_2224_two_fifteen
rgGain_2224_two_sixteen_eq
rgGain_2224_two_sixteen_lt_one_third
not_rgContraction_2224_two_four
rgGain_2224_two_four_eq
rgGain_2224_two_fourteen_eq
rgGain_2224_two_fifteen_eq
rgContraction_two_fourteen_forces_constant_lt_2048
not_rgContraction_two_fourteen_of_ge_2048
not_rgContraction_two_of_ge_2048_of_dmax_le_fourteen
not_rgContraction_2224_two_fourteen
not_rgContraction_2224_two_of_dmax_le_fourteen
rgContraction_of_nonneg_le_2224_two_sixteen
```

Lean proves

$$\texttt{irrelevantScale}\ 2\ 16 = 2^{-13} = 1/8192$$

and the exact gain

$$2224 \cdot 2^{-13} = 139/512 < 1/3.$$

It now also proves the general scale formula for every $d_{\max} \geq 3$:

$$\texttt{irrelevantScale}\ b\ d_{\max} = b^{-(d_{\max}-3)} = \left(b^{d_{\max}-3}\right)^{-1}.$$

For every positive natural block factor b, Lean proves the exact threshold iff

$$\texttt{HasExtendedExtractionContraction}\ C\ b\ d_{\max} \iff 0 \leq C \ \wedge\ C < b^{d_{\max}-3}.$$

Consequently, changing the block factor does not evade the audit; it merely changes the required explicit power threshold. In particular,

$$b^{d_{\max}-3} \leq 2224 \implies \neg\texttt{HasExtendedExtractionContraction}\ 2224\ b\ d_{\max}$$

for $b > 0$ and $d_{\max} \geq 3$. The block-factor-2 specialization is:

$$\texttt{irrelevantScale}\ 2\ d_{\max} = 2^{-(d_{\max}-3)} = \left(2^{d_{\max}-3}\right)^{-1},$$

and the exact iff

$$\texttt{HasExtendedExtractionContraction}\ C\ 2\ d_{\max} \iff 0 \leq C \ \wedge\ C < 2^{d_{\max}-3}.$$

Thus any block-factor-2 contraction certificate at depth $d_{\max} \geq 3$ must place its recombination constant below the explicit power threshold $2^{d_{\max}-3}$. The earlier $d_{\max} = 14$ statement $C < 2048$ is now just the nearest-even-depth specialization of this exact threshold, not an isolated arithmetic calculation. It also proves that the same constant fails to contract at the naive $d_{\max} = 4$ scale, where the factor is $1/2$ and the exact gain is 1112. The nearest even lower extraction depth also fails at the advertised constant:

$$\texttt{irrelevantScale}\ 2\ 14 = 2^{-11} = 1/2048, \qquad 2224 \cdot 2^{-11} = 139/128 > 1.$$

Moreover, any contraction certificate at $d_{\max} = 14$ forces the linear constant to be strictly below 2048, so every proposed $d_{\max} = 14$ certificate with $C \geq 2048$ already fails. Lean then packages the same threshold uniformly for every lower extraction depth $d_{\max} \leq 14$ with block factor 2:

$$2048 \leq C \quad \Longrightarrow \quad \neg\texttt{HasExtendedExtractionContraction}\ C\ 2\ d_{\max}.$$

So the lower-depth repair burden is no longer merely "beat 2224"; any block-factor-2 repair below depth 15 must actually sharpen the recombination constant below the exact threshold 2048. To avoid overclaiming, the audit also records that the raw natural-number cutoff $d_{\max} = 15$ has

$$\texttt{irrelevantScale}\ 2\ 15 = 2^{-12} = 1/4096, \qquad 2224 \cdot 2^{-12} = 139/256 < 1,$$

so the same advertised constant contracts there. The route-level crux is therefore an even canonical-dimension obstruction, not a claim that every natural cutoff below 16 fails. This is not a global Yang–Mills blocker, but it is a precise same-constant obstruction: any repair that keeps the manuscript's advertised combinatorial bound while dropping the even extended extraction depth below 16 must replace this contraction argument with something genuinely different or justify a nonstandard odd cutoff. That residual odd-cutoff loophole is now audited separately in

`Mettapedia/QuantumTheory/YangMills/ExtractionOddCutoff.lean`.

It defines `VanishesOnOddDimensions` and proves

`extendedExtraction_fifteen_eq_fourteen_of_vanishOnOddDimensions`

and

`higherExtractionBlock_fifteen_eq_fourteen_of_vanishOnOddDimensions`.

So on the manuscript's current even-dimensional extracted-operator surface, the putative odd repair $d_{\max} = 15$ is algebraically identical to $d_{\max} = 14$: it adds no new extracted couplings and no new higher block. Therefore the raw natural-number contraction at 15 is not by itself a live repair on that surface; using it would first require genuinely new odd canonical-dimension operators absent from the present extraction model. The RG-facing remainder object is now formalized separately in

`Mettapedia/QuantumTheory/YangMills/ExtractionRemainder.lean`.

This file defines

$$\texttt{extractionRemainder}\ d\ \text{coeff} = \text{coeff} - \texttt{extendedExtraction}\ d\ \text{coeff},$$

proves the pointwise split of coefficients into extracted part plus remainder, and shows that the remainder vanishes exactly when the coefficients are supported up to the cutoff. Most importantly for the odd-cutoff loophole, it proves

`extractionRemainder_fifteen_eq_fourteen_of_vanishOnOddDimensions`.

So on the manuscript's even-dimensional surface, not only the extracted couplings but the actual post-extraction remainder fed into the RG bootstrap is unchanged when one replaces cutoff 14 by 15. The raw 2^{-12} arithmetic therefore does not correspond to a genuinely different remainder being bootstrapped on this surface. That collapse is now pushed to the actual route-witness level in

`Mettapedia/QuantumTheory/YangMills/ExtractionStateRouteCollapse.lean`.

This file defines the downstream RG state

$$\texttt{ExtractionState}\ \alpha := (\texttt{extendedExtraction}\ d\ \text{coeff}, \texttt{extractionRemainder}\ d\ \text{coeff}),$$

proves

`extractionState_fifteen_eq_fourteen_of_vanishOnOddDimensions`,

and then packages the route-facing abstraction

`DeterminedByExtractionState`.

Its manuscript-facing corollaries are

`extractionStateRouteWitness_fifteen_iff_fourteen_of_vanishOnOddDimensions`

and

`not_extractionStateRouteWitness_fifteen_of_not_fourteen_on_oddVanishingSurface`.

So once a proposed odd-cutoff repair witness depends only on the extracted local terms together with the post-extraction remainder, Lean reduces the $d_{\max} = 15$ presentation exactly to the already-blocked $d_{\max} = 14$ surface. The remaining burden is no longer "show that 15 contracts arithmetically"; it is to exhibit some genuinely new downstream datum that is not already contained in this extraction-state pair. That bridge is now pushed all the way back to the existing RG route crux in

`Mettapedia/QuantumTheory/YangMills/ExtractionStateRGCrux.lean`.

This file defines

`ForcesSameConstantFourteenContraction`

for extraction-state predicates P: every cutoff-14 witness of P already yields the manuscript's advertised same-constant contraction `HasExtendedExtractionContraction` 2224 2 14. With that compatibility premise in hand, Lean proves

`sameConstantFourteenGapRoute_of_oddVanishingExtractionStateGapRoute`,

which collapses any odd-vanishing cutoff-15 extraction-state route to the pointed blocked $d_{\max} = 14$ gap surface, and then proves

`sameConstantLowerExtractionGapRouteCertificate_of_oddVanishingExtractionStateGapRoute`

followed by the crux-out theorem

`not_oddVanishingExtractionStateGapRoute_of_forcesSameConstantFourteenContraction`.

So the exact missing compatibility lemma is now isolated in Lean: if an alleged odd-cutoff repair extracts only the same local terms and remainder, and those data already force the manuscript's same-constant $d_{\max} = 14$ contraction, then the whole $d_{\max} = 15$ route is formally impossible.

The manuscript also makes a separate algebraic coherence claim about the two extraction schemes:

$$P_{\leq 4} = P_{\leq 4} \circ P_{\leq d_{\max}}.$$

This is now formalized in

`Mettapedia/QuantumTheory/YangMills/ExtractionProjection.lean`.

The file models a local expansion only at the canonical-dimension coefficient level, via `extractLE`. It does not yet define the actual jet-based local functional projector from the manuscript. On this honest algebraic surface, Lean proves:

```
extractLE_idem
extractLE_compose
extractLE_compose_of_le
extractLE_eq_self_iff_supportedUpTo
standardExtraction_compose_extendedExtraction_of_four_le
```

Thus the manuscript's compatibility identity is now formally available once the actual Yang–Mills extraction operator is shown to act by truncation on the relevant canonical-dimension coefficients. The same file now also isolates the extra extracted couplings above the standard cutoff as `higherExtractionBlock`. On the same coefficient-truncation surface, Lean proves

`standardExtraction_higherExtractionBlock_eq_zero`

and

`extendedExtraction_eq_standard_plus_higherExtractionBlock_of_four_le`.

So for $d_{\max} \geq 4$, the extended extraction splits exactly into the standard $P_{\leq 4}$ part plus the higher $5, \dots, d_{\max}$ block, and that higher block is invisible to the standard extraction. This is still not a construction of the manuscript's jet-based projector, but it sharpens the exact algebra any such projector must satisfy. The same surface is now also explicitly additive:

`extractLE_zero`, `extractLE_add`.

Consequently Lean proves

`standardExtraction_add_higherExtractionBlock_eq`.

So adding a pure higher extracted block to any coefficient family cannot change its $P_{\leq 4}$ projection. At the manuscript level, this means any claimed effect of the extra $5, \dots, d_{\max}$ couplings on the standard extraction/beta-drift channel must come from additional structure beyond this bare truncation algebra. The same coefficient surface now also identifies that higher block exactly as the residual

`higherExtractionBlock_eq_extendedExtraction_sub_standard_of_four_le`,

and Lean proves the equivalent vanishing statement

`standardExtraction_extendedExtraction_sub_standard_eq_zero_of_four_le`.

So after subtracting the standard extraction from the extended one, the remaining $5, \dots, d_{\max}$ residue still has zero $P_{\leq 4}$ projection. The downstream consequence is now packaged separately in

`Mettapedia/QuantumTheory/YangMills/ExtractionObservable.lean`.

It defines observables Φ that are determined by the standard extraction, in the sense that equal $P_{\leq 4}$ projections force equal Φ-values. On that surface Lean proves

`DeterminedByStandardExtraction.extendedExtraction_eq`, `DeterminedByStandardExtraction.add_hig`

and

`DeterminedByStandardExtraction.extendedExtraction_sub_standard_eq_zero`.

So any beta-drift observable or renormalized coupling functional that genuinely depends only on $P_{\leq 4}$ is exactly unchanged by replacing a coefficient family with its extended extraction, by adding the pure higher $5, \dots, d_{\max}$ block, or by passing to the residual $P_{\leq d_{\max}} - P_{\leq 4}$. This makes the route burden explicit: if the manuscript wants the extra extracted couplings to affect the beta channel, that effect has to enter through structure beyond bare dependence on $P_{\leq 4}$. The same file now also isolates the next route-relevant step: if a candidate beta observable G decomposes as a standard-extraction contribution plus an error term E, then all higher-coupling sensitivity is forced into E. Lean proves

`difference_eq_errorDifference_of_add_higherExtractionBlock`, `difference_eq_errorDifference_of`

and

`difference_from_zero_eq_errorDifference_of_extendedExtraction_sub_standard`.

So under the manuscript's own compatibility premise, any claimed effect of the extra extracted couplings on beta drift must be carried entirely by whatever additional higher-order mixing/error term is supplied; it cannot come from the bare $P_{\leq 4}$ contribution itself. This now reaches the manuscript's Remark G.2 language about "the same error bounds." The same file proves

`norm_difference_le_of_add_higherExtractionBlock_errorBound`, `norm_difference_le_of_extendedEx`

and

`norm_difference_le_of_extendedExtraction_sub_standard_errorBound`.

So if the higher-coupling contribution is bounded inside an error term E with some norm estimate, then the full beta-like observable G inherits exactly the same bound. In particular, any advertised $O(\beta_j^{-2})$ control has to be proved at the error-term layer; the $P_{\leq 4}$ part adds no extra dependence on the higher extracted couplings. The same file now also packages the genuinely stepwise drift statement:

`stepDifference_eq_errorStepDifference_of_extendedExtraction`

and its norm corollary

`norm_stepDifference_le_of_extendedExtraction_errorBounds`.

So when one compares the beta-drift increment between two RG steps before and after passing to extended extraction, the entire perturbation is exactly the corresponding perturbation of the error term, and any bound on those two error-term responses transfers directly to the drift increment. The same surface now turns that observation into explicit threshold blockers:

```
abs_difference_le_of_extendedExtraction_errorBound
extendedExtraction_le_target_of_le_target_sub_of_errorBound
extendedExtraction_lt_target_of_lt_target_sub_of_errorBound
stepDifference_le_target_of_le_target_sub_of_errorBounds
stepDifference_lt_target_of_lt_target_sub_of_errorBounds
```

These theorems formalize the route consequence of the manuscript's "same error bounds" language. If a $P_{\leq 4}$-determined real-valued observable is already at least B below a target T, and the extended-extraction error response is bounded by B, then the extended-extracted observable still lies at or below T. Likewise, if a stepwise beta-drift inequality already fails by a margin at least $B_{\mathrm{next}} + B_{\mathrm{cur}}$, then extended extraction cannot repair that failure unless the combined error-term response exceeds that same margin. So a repair route cannot simply reuse the old $O(\beta_j^{-2})$ error budget and claim a threshold or sign correction for free; it must exhibit genuinely larger error response exactly where the standard $P_{\leq 4}$ observable misses the required target.

6 Same-Constant Lower-Extraction Crux

The first dedicated crux-out module is

`Mettapedia/QuantumTheory/YangMills/RGCrux.lean`.

It packages the lower-extraction repair attempt as

`SameConstantLowerExtractionRGCertificate` d_{max}.

This certificate means $d_{\mathrm{max}} \leq 14$ together with the same manuscript linear contraction certificate

`HasExtendedExtractionContraction` $2224\ 2\ d_{\mathrm{max}}$.

Lean proves

`not_sameConstantLowerExtractionRGCertificate`

and the quantified version

`not_exists_sameConstantLowerExtractionRGCertificate`.

Thus there is no same-bound RG certificate at any extraction depth $d_{\mathrm{max}} \leq 14$.

The module also packages the manuscript-shaped version

`SameConstantEvenBelowSixteenRGCertificate` d_{max},

which requires $d_{\mathrm{max}} < 16$, d_{max} even, and the same contraction certificate. The arithmetic lemma

`even_lt_sixteen_le_fourteen`

proves that such a cutoff is at most 14, and hence Lean proves

`not_sameConstantEvenBelowSixteenRGCertificate`

and

`not_exists_sameConstantEvenBelowSixteenRGCertificate`.

This is the preferred statement for comparison with the manuscript's canonical dimensions $4, 6, \ldots, d_{\mathrm{max}}$.

The same file now packages the threshold-level variant

$$\texttt{AtLeast2048EvenBelowSixteenRGCertificate}\ C\ d_{\max},$$

which requires $d_{\max} < 16$, $d_{\max}$ even, $2048 \le C$, and the same block-factor-2 contraction certificate. Lean proves

`not_atLeast2048EvenBelowSixteenRGCertificate`

and

`not_exists_atLeast2048EvenBelowSixteenRGCertificate`.

So the obstruction is no longer tied only to the manuscript's literal 2224: any lower-even extraction repair that keeps the recombination constant at or above 2048 already fails before any clustering or Osterwalder–Schrader argument is reached.

The same obstruction is lifted to the surrounding mass-gap route certificate:

`SameConstantLowerExtractionGapRouteCertificate`.

This predicate includes a candidate spectral gap, a spectral-gap-to-clustering bridge, and a lower-extraction same-constant RG certificate. Lean proves

`not_sameConstantLowerExtractionGapRouteCertificate`

and, for the even manuscript-shaped route certificate,

`not_sameConstantEvenBelowSixteenGapRouteCertificate`.

The strengthened threshold version is also lifted as

`AtLeast2048EvenBelowSixteenGapRouteCertificate`

with theorem

`not_atLeast2048EvenBelowSixteenGapRouteCertificate`.

and the pointed $d_{\max} = 14$ form

`not_sameConstantFourteenExtractionGapRouteCertificate`.

This theorem deliberately does not refute a Yang–Mills mass gap. It refutes one repair corridor for the manuscript route: keeping the advertised constant 2224, using block factor 2, and lowering the extended extraction depth below 16. A viable repair must either sharpen the $d_{\max} = 14$ constant below 2048, keep $d_{\max} = 16$, or replace the linear contraction certificate with a genuinely different argument.

7 Regression

The companion module

`Mettapedia/QuantumTheory/YangMills/MassGapRegression.lean`

contains theorem-level wrappers for the positive-gap projection, pointwise spectral obstruction, downward closure, positive-energy spectral splitting, and quadratic-form monotonicity. It also checks the threshold/subset forms of the spectral exclusion, including the positive-energy non-vacuum threshold and the finite-mass consequences, including the bounded-above characterization, the least-upper-bound API for `SpectralGapSup`, its positivity consequence, the positive-spectrum upper bound, the first-positive-spectral-value exact endpoint equivalence, uniqueness, gap-set/supremum identifications, and the positive-spectrum infimum comparison, and the unbounded-gap obstruction.

The companion clustering regression module

`Mettapedia/QuantumTheory/YangMills/ClusteringRegression.lean`

pins the fixed-rate projection, monotonicity in the endpoint Δ, the rate-level violation blocker, the bridge projections from spectral gap to clustering, the final contrapositive obstruction from a clustering-rate failure back to failure of the proposed spectral mass gap, the quantitative gap upper-bound form, the supremum upper-bound form, and the all-positive-rates no-gap criterion. It also pins the positive-long-range-lower-bound route blocker and its conversion into persistent fixed-rate clustering violations.

The RG bootstrap regression module

`Mettapedia/QuantumTheory/YangMills/RGBootstrapRegression.lean`

pins the exact irrelevant-scale values for $d_{\max} = 4, 14, 15, 16$, the general reciprocal-power threshold for every b and $d_{\max} \geq 3$, the exact positive-block-factor contraction iff $0 \leq C$ and $C < b^{d_{\max}-3}$, the advertised-constant blocker when $b^{d_{\max}-3} \leq 2224$, the block-factor-2 specialization, the advertised extended-extraction contraction, the raw $d_{\max} = 15$ contraction, the exact gain values $139/512 < 1/3$, $139/256$, $139/128$, and 1112, the naive $d_{\max} = 4$ noncontraction, the nearest-lower-even-depth $d_{\max} = 14$ noncontraction and threshold $C < 2048$, the stronger blocker for every $C \geq 2048$ at $d_{\max} = 14$ and, more generally, at every $d_{\max} \leq 14$, the monotone variant for any nonnegative constant bounded by 2224, the invariant-interval bootstrap lemma, and the finite-envelope/geometric-zero-error wrappers.

The extraction observable regression module

`Mettapedia/QuantumTheory/YangMills/ExtractionObservableRegression.lean`

pins the factorization criterion through $P_{\leq 4}$, the invariance under added higher extracted blocks and extended extraction, the vanishing of the extended-minus-standard residue, the pure-error identities for higher-block, extended-extraction, and residual responses, the inherited norm bounds from the error term, and the step-difference/error-difference identity together with its norm bound. It now also pins the real-valued threshold forms: absolute observable perturbation is bounded by the same error budget, any observable already below a target by more than that budget stays below the target after extended extraction, and the same quantitative obstruction holds for step-drift inequalities using the combined two-step error response.

The odd-cutoff regression module

`Mettapedia/QuantumTheory/YangMills/ExtractionOddCutoffRegression.lean`

pins the successor-cutoff collapse when the new coefficient vanishes, the odd-successor collapse on odd-vanishing coefficient families, and the manuscript-facing specializations `extendedExtraction` $15 =$ `extendedExtraction` 14 and `higherExtractionBlock` $15 =$ `higherExtractionBlock` 14 on that even-dimensional surface.

The extraction remainder regression module

`Mettapedia/QuantumTheory/YangMills/ExtractionRemainderRegression.lean` pins the coefficient decomposition into extracted part plus remainder, the pointwise vanishing/agreement behavior below and above the cutoff, the exact characterization of zero remainder by supportedness up to the cutoff, and the manuscript-facing equality of the actual remainders at cutoffs 15 and 14 on odd-vanishing coefficient families.

The extraction-state route-collapse regression module

`Mettapedia/QuantumTheory/YangMills/ExtractionStateRouteCollapseRegression.lean`

then pins the full extraction-state equality at cutoffs 15 and 14, the generic route-witness collapse for predicates factored through that state, a positive odd-vanishing example where the witness does collapse, and a negative odd-sensitive example showing that the odd-vanishing hypothesis is genuinely used.

The extraction-state RG-crux regression module

`Mettapedia/QuantumTheory/YangMills/ExtractionStateRGCruxRegression.lean`

then pins the route-level collapse into the pointed $d_{\max} = 14$ gap surface, the induced lower-extraction gap certificate, the generic no-go theorem, and a concrete negative example showing that not every arbitrary extraction-state predicate can force the blocked cutoff-14 contraction.

The RG crux regression module

`Mettapedia/QuantumTheory/YangMills/RGCruxRegression.lean`

pins the nonexistence of same-constant lower-extraction RG certificates, the nonexistence of such certificates at any $d_{\max} \leq 14$, the even-cutoff lemma reducing $d_{\max} < 16$ and even to $d_{\max} \leq 14$, and the lifted route-certificate blockers for the surrounding mass-gap/clustering bridge surface. It now also pins the threshold-level nonexistence of every even-cutoff lower-extraction certificate with recombination constant at least 2048, together with the corresponding lifted route blocker.

8 What Remains

This layer does not solve the Clay problem. The hard work still includes:

- constructing a nontrivial four-dimensional quantum Yang–Mills theory;
- choosing a mathematically adequate gauge group and field-strength formalization;
- connecting bounded-operator placeholders to the physically relevant unbounded Hamiltonian theory;
- proving a genuine spectral gap for that Hamiltonian;
- relating the spectral gap to clustering and local operator estimates.
- instantiating `SpatialCorrelation` with centered Yang–Mills local-operator vacuum correlations and proving the polymer/tree-decay estimates needed by the manuscript's OS reconstruction route.
- instantiating `AffineRGStepBound` with the actual polymer norm recurrence, including the one-step RG estimates and all constants feeding the advertised $2224 \cdot 2^{-13} < 1$ contraction.

- for any lower-even-extraction repair with $d_{\max} < 16$, either sharpening the $d_{\max} = 14$ recombination constant below 2048, justifying a nonstandard odd cutoff such as $d_{\max} = 15$, or abandoning the same linear contraction certificate.

The value of the current step is that future claims now have a precise Lean interface for the mass-gap and clustering audit obligations.